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COLLEGE OF INFORMATICS AND COMPUTING INNOVATIONS (CICI)
Tiwi Campus, Barotac Nuevo, Iloilo | Capstone Project Research Group

SECUREFACULTY: FULL SYSTEM DEPLOYMENT GUIDE

Standard Local Production Setup: XAMPP (Apache + MySQL) & Python Engine (No Docker)

DOCUMENT: Host Deployment Manual | **WEB STACK: Apache 2.4 + PHP 8.2 (XAMPP)** | **DATABASE: MySQL / MariaDB (InnoDB)**

Operational Scope: This guide outlines the exact, foolproof procedure to install and run the complete SecureFaculty systematic records management portal directly on a client machine using standard **XAMPP (Apache & MySQL)** and **Python 3.10+**. No Docker or virtualization is required. All data is persisted in a high-performance relational MySQL/MariaDB database with strict foreign keys, multi-author indexes, and AES-256 vault encryption.

1. Software Requirements & Exact Versions Needed

Download and install the following two components on the client computer:

Software Component	Exact Tested Version	Included Modules	Official Download Source
XAMPP for Windows	**8.2.12** (or 8.1.x – 8.2.x)	Apache 2.4.58, PHP 8.2.12, MariaDB 10.4.32, phpMyAdmin	apachefriends.org/download.html
Python for Windows	**3.11.x** or **3.12.x** (64-bit)	pip, Pillow, PyPDF, python-docx, ReportLab 4.0+	python.org/downloads/windows

CRITICAL INSTALLATION FLAG: When installing Python, you MUST check the box: "Add python.exe to PATH" at the bottom of the first setup window. This enables PHP and the web server to invoke the conversion engine.

2. Project Directory Placement

Copy or clone the entire SecureFaculty project folder into the XAMPP web root directory:

```
Target Directory:   C:\xampp\htdocs\securefaculty
Ensure folders 'storage\vault' and 'storage\temp' exist inside this directory.
```

3. PHP Configuration (php.ini)

Open **C:\xampp\php\php.ini** (or click **Config -> PHP (php.ini)** in XAMPP Control Panel):

```
// 1. Verify these extensions are enabled (remove the leading ';' semicolon if present):
extension=pdo_mysql      # Required for MySQL/MariaDB database
extension=openssl        # Required for AES-256-CBC cryptographic vault
extension=fileinfo       # Required for server-side MIME type validation
extension=curl           # Required for OCR.space cloud API failover
extension=mbstring       # Multibyte character encoding
extension=gd             # Image thumbnail and preview support
// 2. Set upload limits for large academic manuscripts & certificates:
upload_max_filesize = 50M      post_max_size = 50M      memory_limit = 256M
```

4. Apache URL Rewrite Configuration (httpd.conf)

In **C:\xampp\apache\conf\httpd.conf**, ensure mod_rewrite is enabled for clean URLs:

```
// Ensure mod_rewrite is enabled (uncomment line by removing '#'):
LoadModule rewrite_module modules/mod_rewrite.so
// In Directory "C:/xampp/htdocs", verify AllowOverride is set to All:
AllowOverride All
```

5. MySQL Database Setup & Schema Import (MariaDB)

Import the pre-configured database schema with bcrypt credentials and sample research publication:

Step 5.1 Open **XAMPP Control Panel** and click **Start** for both **Apache** and **MySQL**.

Step 5.2 Open browser and go to: **http://localhost/phpmyadmin/**

Step 5.3 Click the **Import** tab in the top navigation bar.

Step 5.4 Click **Choose File**, select **C:\xampp\htdocs\securefaculty\sql\schema.sql**, and click **Import** (or **Go**).

** CLI Alternative: Open CMD and execute: C:\xampp\mysql\bin\mysql -u root < C:\xampp\htdocs\securefaculty\sql\schema.sql*

SECUREFACULTY: ACCESS, CREDENTIALS & VERIFICATION

Python Dependencies, Portal URLs, Demonstration Accounts, and System Checks

6. Install Python Dependencies (Run Once in Terminal)

Open Command Prompt or PowerShell inside **C:\xampp\htdocs\securefaculty** and run:

```
pip install -r requirements.txt
# Or manually: pip install pillow>=10.0 pypdf>=4.0 python-docx>=1.0 requests>=2.28 reportlab>=4.0
```

7. Accessing SecureFaculty in Web Browser

Once Apache and MySQL are running in XAMPP, open your browser and navigate to:

Official Application URL: http://localhost/securefaculty/ (or http://127.0.0.1/securefaculty/)

8. Pre-Configured Demonstration Accounts (RBAC)

The database comes pre-seeded with Dean administrative clearance and faculty test profiles:

Role / Clearance	Username	Password	Full Name & Position	Permissions & Workspace Scope
Dean (Admin)	`dean`	`admin123`	Dr. Arlene S. Poblador *Dean, CICI*	Full clearance, administrative dashboard, user registry, compliance audit ledger, department vault
Faculty Member	`mikhail`	`faculty123`	Prof. Levy Mikhail B. Aguilar *Assistant Professor II*	Personal encrypted vault, document ingestion gateway, primary author on Capstone publication
Faculty Member	`raejan`	`faculty123`	Prof. Rae Jan B. Corneja *Instructor I*	Personal encrypted vault, co-author attribution badges on joint research publication
Faculty Member	`eljay`	`faculty123`	Prof. Eljay Muyco *Instructor I*	Personal encrypted vault, co-author attribution badges on joint research publication

9. Key Feature Demonstration Checklist

Test these core workflows to demonstrate the system during capstone defense or client review:

Smart Pipeline	Digital Vault (AES-256)	Audit Trail (RA 10173)	Accreditation Report
Click **Ingestion Pipeline** -> 'Load Demo Sample' -> Stage 1 to 4 telemetry displays OCR extraction & H-I-T-L validation.	Zero plaintext on disk. Clicking 'Decrypt & View' recovers payload directly in volatile RAM.	Log in as Dean -> **Audit Trail**. Immutable compliance ledger tracking IPs, users, and actions.	Click **Accreditation Report** -> PDF. Generates publication-grade A4 Portrait PDF report with letterhead.

10. Troubleshooting Quick Reference

- **Apache Port 80 / 443 Conflict:** If Apache fails to start, click **Config -> Service & Port Settings** in XAMPP and change the Main Port to 8080. Access via http://localhost:8080/securefaculty/.
- **MySQL Port 3306 Conflict:** If another MySQL service is running, stop it via Windows Services (services.msc) or use the XAMPP MySQL service.
- **'python' command not recognized:** Re-run Python installer, select *'Modify'*, and check *'Add Python to environment variables'*.
- **HTTP 404 on clean URLs (/dashboard, /vault):** Verify in httpd.conf that LoadModule rewrite_module is enabled and AllowOverride All is set.

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